X-57 Maxwell

Flight Manual

Mod III and Mod IV, for FlightGear with the JSBSim flight model



This manual describes the aircraft as simulated. Mods III and IV never flew, and NASA's own Pilot's Operations Handbook isn't public, so this is not a real aircraft's manual and its procedures are for the simulator only. It comes from NASA's published Cockpit Interface Control Document and papers and the Tecnam P2006T flight manual. This work is not connected with or endorsed by NASA or Tecnam.

Generated from the aircraft's sources (`manual/` in its repository). November 2026.

How to use this manual

The manual runs as a flight manual does: the aircraft, its cockpit and systems, limitations, normal procedures, flight characteristics, emergency procedures and performance. Appendix A sets the cockpit beside NASA's handbook, and lists every place the two differ. Keys are given as g; words on the panel, as `ESS BATT`.

Performance figures are the model's, with NASA's predictions beside them where there are any. How far to trust each one is in the model's report, *Flight Dynamics of the X-57 Maxwell in JSBSim*.

Warnings, cautions and notes. Three kinds of notice stand apart from the text, by how much they matter:

WARNING

A procedure or technique that, not followed, may lose the aircraft.

CAUTION

A procedure or technique that, not followed, may damage the aircraft or end the flight's purpose.

Note

A procedure, technique or fact worth emphasising.

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Chapter 1

General description

1.1 The aircraft

The X-57 Maxwell is NASA's battery-electric demonstrator, built on a Tecnam P2006T. Its forms are called Mods:

Mod II the P2006T with its piston engines replaced by electric motors. Built and tested on the ground. Not simulated here.

Mod III a new wing of 42 % of the area and aspect ratio 15, with the cruise motors at the tips, where their propellers turn against the tip vortices. The high-lift motors are on it, folded and inactive. The wing was built and tested; the aircraft never flew.

Mod IV Mod III with twelve folding high-lift propellers along the leading edge, to fly slowly on so small a wing. Never built.

From Mod II on, a flight-test air-data boom of the NACA style projects from the nose's right side, its pitot tip about 1.2 m ahead; its two vanes, for angle of attack and sideslip, trail into the airflow in flight. Its sizes are estimates from NASA's rendering. Both Mods share this cockpit. On Mod III the high-lift panel stays dark. The cabin is NASA's: one seat, the pilot's. The copilot's seat, yoke and pedals gave way to a research instrumentation pallet, which holds the data logger, and the rear seats to the traction battery modules and their control systems. Their look follows NASA's photographs; their sizes and places are estimates. The pilot's rudder pedals, with toe brakes, swing with the rudder and brakes. Outside the Cockpit View a pilot sits in the left seat, his hands on the yoke and torque levers and his feet on the pedals, following them. He is 1.75 m tall, on a seat lowered 8 cm from the P2006T model's place, as the real seat adjusts, and he wears its three-point seatbelt and black work shoes. He wears a headset and green-lensed aviator sunglasses.

Table 1.1: Principal data.

	Mod III	Mod IV
Wing area, span	66.7 ft^2, 31.6 ft	
Weight flown in FlightGear	3000 lb	3200 lb (projected)
Centre of gravity	28.6 % of the chord: no fuel, so it stays put	
Cruise propellers	two, 3 blades, 1.52 m, electric variable pitch	
Cruise motors	255 Nm, 72 kW at 2700 rpm, two controllers each	
High-lift propellers	twelve, folded, inactive	twelve, 5 blades, 0.58 m, folding, fixed pitch
Traction battery	two packs, 5120 cells in all, 51.2 kWh usable	
Target cruise	150 KTAS at 8000 ft	

1.2 Controls in FlightGear

Key	Action
Page Up / Page Down	Torque levers
G	Landing gear
V	Cycle the views, among them the Battery View (the aft cabin's batteries, their control modules and vent) and, after it, the Pilot Face View (the pilot's face, headset and sunglasses, from over the yoke)
[and]	Flaps
H	High-lift ARM or DISARM/RESET
Y	Yoke disarm (or click the red button on the yoke)
Click a door	Opens or shuts it, from inside or out: the main door on the left, the rear door on the right. On the ground only, below 19 kt. Hover over a door to see whether it is open and whether a click will work. Each door clicks open and thumps shut, and while one is open the cockpit hears the outside

The levers use FlightGear's throttle and propeller pitch controls.

- **X-57 menu** > **Autostart** runs NASA's start-up order in 25 seconds, needed only after shutting down.
- **X-57 menu** > **Autostop** runs the shutdown, on the ground only.
- **X-57 menu** > **Systems and failures** sets NASA's failures and the levers' overdrive and regeneration.

Chapter 2

The cockpit



Figure 2.1: The pilot's view.



Figure 2.2: The instrument panel.

2.1 Left panel



Figure 2.3: Left panel.

Control	What it does
Annunciator	See below
AUDIO ANN. ACK	Silences the current aural alerts
AUDIO CHECK	Tests the aural alert while held
LIGHT CHECK	Tests the annunciator lights while held
Airspeed, attitude, altimeter, turn, heading, climb	Stock P2006T. Attitude, heading and turn are on DC Converter Bus A
UHF radio	COM 1, on the Essential Bus
G-meter (GT-50)	Mode steps through g, Essential Bus volts, timer, flight timer and clock. Action shows maximum and minimum g, or starts the timer. Reset clears them
Trim indicators	Stabilator and rudder

2.1.1 High-lift panel (Mod IV)

Control	What it does
ARM, DISARM/RESET	Arms or disarms all twelve motors. Moving to DISARM/RESET also clears faults and a yoke disarm
AIRSPEED, FIXED	Motor speed from the airspeed schedule, or fixed. You can change it without disarming
RESET REQ'D (yellow)	The yoke switch was used. Move ARM to DISARM/RESET
RDY (blue)	Steady: ready. Flashing: starting up or waiting for high voltage
RUN (green)	Steady: running. Flashing: spinning up or down
FAULT (red)	Steady: armed with no high voltage. Flashing: fault while armed

2.1.2 Annunciator

Top row	Bottom row	Colour	Lights when
DC CONV A	DC CONV B	Yellow	Converter failed, or traction battery below 310 V
MAIN BATT A	MAIN BATT B	Red	Cell above 65 °C
TRAC BUS A	TRAC BUS B	Yellow	Traction bus at or below 350 V
LEFT CMC A	LEFT CMC B	Yellow	Controller failed, or armed with no traction voltage
RIGHT CMC A	RIGHT CMC B	Yellow	The same
MAIN DOOR OPEN	REAR DOOR OPEN	Red	Door open
PITOT HEAT (green)	PITOT HEAT (yellow)		Not modelled
GEAR PUMP ON		Yellow	Gear moving
	ESS BATT	Red	Battery failed (steady), or below 12.8 V (slow flash)

Red means act now. Yellow means caution, and return to base if it can't be cleared.
The aural alert repeats every two seconds until acknowledged with AUDIO ANN. ACK.

2.2 Centre stack



Figure 2.4: Centre stack.

Control	What it does
Crew panel (Mod IV)	Blue, green and red lights for each of the twelve high-lift motors. The right switch turns the display on. The left switch is a continuity check, with no effect here
Navigator	Shows COM 2 and NAV 1 frequencies only
Transponder	Shows the code
Page knob	D175 page. 1 is the pilot's page, the others are secondary
FAULT/MAIN	Stays at MAIN. It no longer does anything on the aircraft either
START (green)	Blades at the fine stop. On the aircraft it means the end of travel
FEATHER (blue)	Feather is commanded, not that the blades have arrived
LH FAULT, RH FAULT (yellow)	Propeller hub fault
D175	See below
Audio panel	Not modelled

2.2.1 D175 pages

Page	Shows
1	Torque lever %, estimated torque, motor power, battery volts, state of charge
2	Controller temperatures (dashes: not modelled)
3	Motor temperatures (dashes)
4	Software versions and propeller pitch
5 to 7	Windings and nacelle temperatures (dashes)

The ten lights along the top show CAN bus traffic. From left: left data acquisition, left CMC A and B, left BMS, left torque encoder, right torque encoder, right BMS, right CMC A and B, right data acquisition.

CAUTION

NASA treats the D175 as test information only. Trust the energy meters for the battery's state: NASA's state of charge drifts.

2.3 Right panel



Figure 2.5: Right panel.

Control	What it does
Tachometers	Propeller rpm. Digits from 300 rpm, bar green from 500, red above 3,080. Left on Bus A, right on Bus B
Traction bus meters	Volts, amps and power for each controller. They autoscale: past 1,000 W the power reads in kW with only a small mark at the left, as on the aircraft
Avionics meters	Volts and amps of DC converter A and B. Bar red outside 13 to 14.6 V or 0 to 70 A
MAN/AUTO	AUTO: the propeller lever sets rpm, from 1,700 to 2,700. MAN: blades move only while you hold START or FEATHER
START, FEATHER tags	Click and hold. START drives to fine pitch, FEATHER to coarse. In MAN only
OAT	Outside air temperature, °C

2.4 Overhead and levers



Figure 2.6: Overhead panel: the traction battery and contactors.



Figure 2.7: Pedestal: torque and propeller levers.

Lever	Position	Effect
Torque	REGEN	Below idle. Negative torque, recharging the battery
	IDLE	8 Nm per controller
	FULL	127.5 Nm per controller, 255 Nm per motor
	OVERDRIVE	Up to 175 Nm, for one controller left. Past a detent
Propeller	Forward	2,700 rpm
	Cruise	2,250 rpm
	Low	1,700 rpm
	Aft stop	FEATHER: drives the blades to feather, overriding MAN and AUTO
Parking brake	Out	Set: holds the brakes on. A round knob at the foot of the pedestal, as on the P2006T: click it, or press B. Set when the aircraft loads. The brakes, set or pedalled, hold the aircraft to about half torque at 2700 rpm: more and it slides
	In	Released

FlightGear's throttle stops at idle and full, so tick Overdrive and Regeneration in **X-57 menu** > **Systems and failures** to reach those regions.

2.5 Lower panels

Control	What it does
Gear lever	Up or down. Gear runs on the Essential Bus. On the ground the gear stays down whatever the lever says, and comes up as soon as the wheels leave the ground if the lever is up. That guard is a simulator choice: sources disagree on whether the real aircraft has a squat switch
NOSE, LH, RH (green)	Gear down and locked
TRANS (red)	Gear moving
LT TEST	Tests the gear lights while held
Flap switch and indicator	As on NASA's Mod III and IV, the flaps run while the switch is held and stop where it's let go: press and hold just above the lever to raise them, just below it to lower them, and the lever springs back to the middle. They move 3 degrees a second. The scroll wheel over the lever nudges them 2.5 degrees; [and] step 0, 10, 20 and 30 from wherever they stand. The dial reads in degrees: NASA takes off with 10 and lands with 30. Started on a runway, they're already at 10; at a parking position or in the air, up. NASA's data has flaps 0, 10 and 30, so in between the model interpolates
LANDING, TAXI, NAV, STROBE	The exterior lights, all on DC Converter Bus B in NASA's power drawings: 3 A breakers, 7.5 A for the strobe. Those drawings collar out landing, taxi and nav, the one with the Mod IV additions included, so only the strobe was to be live. Here all four work. They don't say where the lights sit: the tip nacelles, the tail and the fin are this model's choices. Landing and taxi are in the left wing's leading edge, where NASA's photographs show them on Mod II's stock wing. No source shows them on this wing, so they keep the same distance from the centreline
INSTR (fitted only)	Not on the X-57: X-57 menu > Tecnam cockpit lights fits the P2006T's. The fifth rocker in the LIGHTS row, as on the P2006T: the instrument lights, on DC Converter Bus B through the INSTR LIGHT breaker, which NASA's drawings collar out. Three dimmers appear under the EXPERIMENTAL placard: switch lights, avionics key lights, and the two ceiling cockpit lights over the left and right panels. The cabin light and the emergency light are in the ceiling, each with its switch on the lamp. Unfitted, none of it is drawn
CABIN FAN	No effect
DAU ENABLE	Data system. Green Time Lock light
RF ENABLE	Telemetry. No effect here
RECORD	Light: green ready, red archiving, amber fault

2.6 Breaker panels

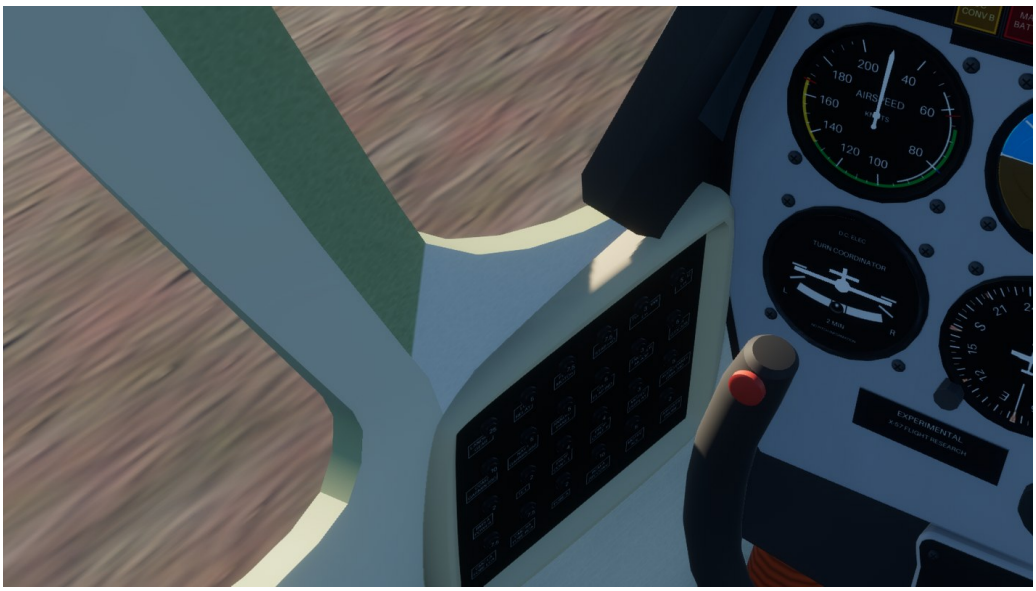


Figure 2.8: Left breakers, from NASA's photograph, in their cream moulding between the instrument panel and the main door's frame.



Figure 2.9: Right breakers: names from NASA's power drawing, their order a guess.

Click a breaker to pull it. Four do something: LANDING GEAR, FLAP MOTOR and PROP CONT. L and R.

Chapter 3

Systems

3.1 Propulsion

Each cruise motor has two controllers, A and B. The torque lever commands torque: 127.5 Nm a controller at FULL, so 255 Nm a motor. The propeller lever sets rpm through an electric motor in the hub, which turns the blades at only 1 degree a second. In AUTO the hub governs above 200 rpm; in MAN the blades move only while START or FEATHER is held.

3.2 High lift (Mod IV)

Twelve motors drive folding propellers, in four groups, A and B on each wing, each with its own contactor. They do nothing unless selected. All six on a wing turn the same way as its cruise propeller, inboard blade rising, and the left wing mirrors the right.

FIXED 4800 rpm, for takeoff.

AIRSPEED the speed follows the airspeed schedule, for landing. Below 95 KEAS the torque rises from 3 Nm towards 22 Nm at 58 KEAS.

Disarmed, the blades fold along the nacelles.

3.3 Electrical power

Bus	Fed by	Carries
Essential Bus	Essential battery, cross buses	Gear, flaps, propeller control, annunciator, D175, G-meter, radios, contactor power
DC Converter Bus A	Converter A	Attitude, turn and heading, left tachometer
DC Converter Bus B	Converter B	Right tachometer, rudder trim, strobe, cabin fan
Wing Avionics A, B	Essential Bus at 23 V	Controllers LA, RA and high-lift A; LB, RB and high-lift B
Avionics Bus A	DC Converter Bus A	Transponder
Avionics Bus B	DC Converter Bus B	Instrumentation

The Essential Bus is the last one to lose: with it dead the contactors open and all traction goes. To shed load, turn off AVIONICS BUS A and B first.

3.4 Exterior lights

NAV lights the tip nacelles and tail, STROBE flashes the tips and fin top, and LANDING and TAXI are in the left wing's leading edge, all on DC Converter Bus B. Position lights and strobes start on.

3.5 Cockpit lights

The X-57 has none. NASA's power drawings keep the P2006T's INSTR LIGHT circuit, 5 A on DC Converter Bus B, collared out; its photograph of the cockpit has blanks where the P2006T has its dimmers, and its own panel where the P2006T's overhead console was. Its flights were to start within an hour of sunrise, and none was flown: the aircraft never reached a full flight configuration.

For flying in the dark, **X-57 menu > Tecnam cockpit lights (not on the X-57)** fits the lights a P2006T has. Until then nothing of them is in the cockpit. Fitted, they are:

- INSTR, a fifth rocker in the LIGHTS row, as on the P2006T: the instrument lights, on DC Converter Bus B through the INSTR LIGHT breaker.
- Three dimmers at the P2006T's place for them, which on the X-57 holds its EXPERIMENTAL placard: fitted, they take the placard's place. From the left: the switches' built-in lights, the avionics' key lights, and the cockpit lights, two lamps in the ceiling that light the left and right panels. Drag a knob up or down.
- The cabin light in the ceiling behind them, on the Essential Bus, its switch on the lamp.
- The emergency light, five LEDs just aft of the traction power panel, straight from the Essential battery, its switch on the lamp.

The places come from the P2006T's flight manual and a model of its overhead console; the levels are estimates. The same switches and dimmers are in the menu's window.

3.6 Landing gear and flaps

The gear is the P2006T's, electro-hydraulic, and takes about 6 seconds. It runs on the Essential Bus. TRANS shows it moving, and GEAR PUMP ON the pump running. The flaps have one electric motor on the Essential Bus, at 3 degrees a second: 10 for takeoff, 30 for landing. On the ground the gear stays down whatever the lever says, and comes up as the wheels leave the ground if the lever is up. That guard is a simulator choice: sources disagree on a squat switch.

The pedals move the rudder, 26 degrees each way, and steer the nose wheel with it in proportion, to 20. The linkage has some play: within 2 degrees either side of where the pedals put it, the nose wheel trails. How much play the real aircraft has isn't published; this figure is fitted to NASA's ground simulation. At speed the nose wheel bites hard. In a 15 kt crosswind from behind, about 3 degrees of nose wheel holds the line; 5 or more above 20 kt leans the aircraft onto a wingtip propeller.

Chapter 4

Operating limitations

Table 4.1: Limits.

Never exceed speed	172 KCAS
Propeller speed	1700 to 2700 rpm, tachometer red above 3080
Torque	255 Nm a motor; 175 Nm overdrive on one controller, with the other lost
High-lift deployment	select AIRSPEED above 120 KEAS, ARM between 120 and 95 KEAS
Traction bus	caution at or below 350 V
Traction battery	caution below 310 V; cell above 65 °C is a red alert
Essential battery	flashing alert below 12.8 V
Converter output	13 to 14.6 V, 0 to 70 A
Ceiling	14 000 ft capable, 10 000 ft operational
Flap settings	0, 10, 30 degrees

Note

No gear or flap speed limit is published for the X-57, and the model has none. The P2006T's are 93 KIAS, which the X-57's own takeoff speeds exceed.

Chapter 5

Normal procedures

5.1 Start-up

Everything is on the panel. Each low voltage switch goes up, working from left to right.

Step	Control	What happens
1	ESS BATT	Essential battery powers the Essential Bus. Annunciator, D175, G-meter and radios come on
2, 3	DC CONV A, B	Each 13.8 V converter comes online. Needs the Essential Bus
4, 5	CROSS BUS A, B	Converters feed the Essential Bus, share the load and charge the battery
6, 7	WING AVIONICS A, B	23 V to the motor controllers and high-lift controllers
8, 9	AVIONICS BUS A, B	Transponder and data system
	DAU ENABLE, RF ENABLE, RECORD	Data system, if wanted
10, 11	MAIN BATT A, B (overhead)	Traction battery contactors close
12	LA, LB, RA, RB (overhead)	High voltage to each controller. Wait for each LED to turn green
	HIGH-LIFT BUS A, B (overhead, Mod IV)	High voltage to the high-lift controllers. H-LA to H-RB turn green
13	RIGHT CMC A, B, then LEFT CMC A, B	Arm the controllers: the motor turns at idle. The right side goes first, to keep the left door clear

CAUTION

Before starting a motor the blades must be at START, because AUTO only works above 200 rpm. If the green START light is out, set MAN and hold START.

5.2 Takeoff

5.2.1 Speeds

Speed (KIAS)	Mod II	Mod III	Mod IV with DiTTo
Stall, Vs1	60	80	70
Rotate		88	77
Initial climb, Vy	72	96	84
Minimum control, Vmca	60	86	73

5.2.2 Mod III

Flaps 10, full torque, 2700 rpm. The P2006T's brakes hold the aircraft only to about half torque, so let them off and push the levers up as it rolls. Rotate at 88 KIAS and climb at 96. Be ready for a long run: about 4300 ft with the model's automated test pilot. Raise the gear and flaps at a safe height.

5.2.3 Mod IV: Distributed Thrust Takeoff (DiTTo)

DiTTo is a technique for the pilot, not software. Nothing automatic mixes the motors or reacts to the rudder and ailerons. Holding the wingtip motors to about two thirds power keeps a motor failure controllable, and the high-lift motors make up the thrust.

1. During the run-up at raised torque, set the propeller levers to 1800 rpm and leave them there.
2. Line up, select FIXED and move to ARM. All twelve high-lift motors spin up to 4800 rpm. RDY goes out and RUN turns green. Their thrust alone, about 750 lbf, is more than the model's brakes hold, so it starts rolling: arm on the runway, lined up.
3. Push the torque levers to the FULL detent, the top of the throttle unless Overdrive is ticked. That's 255 Nm and about 48 kW a motor, 67 % of rated power.
4. Rotate at 77 KIAS and climb at 84 KIAS.
5. At a safe height, accelerate to best rate of climb, then raise gear and flaps and disarm the high-lift motors.
6. Cruise climb at 2250 rpm and 255 Nm, about 60 kW a motor.

5.3 Cruise

At 2250 rpm, trim and set torque for the speed. 150 KTAS at 8000 ft takes about 85 kW in the model. Watch the energy meters, not the D175's state of charge.

5.4 Approach and landing

5.4.1 Mod III

With no high lift the approach is fast: NASA's reference speed is 94 KCAS. Gear down, flaps 30.

5.4.2 Mod IV

1. Select AIRSPEED while above 120 KEAS.
2. ARM between 120 and 95 KEAS.
3. Gear down, flaps 30, and fly 75 KCAS. The high-lift propellers now hold the wing up: the model settles at about 4300 rpm and 5 kW a disc. Their thrust works against the descent, so the torque levers stay near idle.
4. After touchdown, press the red disarm button on the yoke's left horn. The propellers stop and the blades fold. Until then their thrust outruns the brakes: NASA's pilots found that in the simulator, and the button was added for it.
5. On a go-around, disarm above 120 KEAS and the blades fold.

WARNING

On Mod IV at approach speed the wing is flying on the high-lift propellers. Losing them all takes the stall speed from 58 KCAS to about 75, Mod III's 73 at this weight: that's the approach speed.

5.5 Shutdown

Torque levers to idle, disarm left then right, open the contactors, then MAIN BATT. Then work the low voltage switches from right to left, with ESS BATT last.

Chapter 6

Flight characteristics

6.1 Modes

Table 6.1: Mod III's modes in the model, NASA's simulator in brackets.

	88 KCAS	96 KCAS	112 KCAS
Phugoid damping ratio	0.099 (0.104)	0.104 (0.116)	0.120 (0.122)
Short period damping ratio	0.361 (0.374)	0.414 (0.382)	0.467 (0.439)
Dutch roll damping ratio	0.106 (0.118)	0.124 (0.136)	0.157 (0.158)
Roll time constant, s	1.00 (1.03)	0.81 (0.83)	0.62 (0.66)

The aircraft is lightly damped in yaw, and the roll is slow for its size: the long wing's weight is far out, with a motor at each tip. Near stall the Dutch roll gets much weaker: its damping falls to 0.04 at 80 KCAS in the model (NASA's simulator 0.11).

6.2 Stall

Mod III stalls fast, and with little to spare: at 3000 lb the model's trimmed stall speeds are 82.7, 80.6 and 70.8 KCAS with flaps 0, 10 and 30, NASA's 88, 80 and 73. The high-lift propellers on Mod IV bring it down to 58 KCAS.

6.3 The wing

The wing bends in its first mode, about 2.4 Hz up and down together and 3.5 to 3.9 Hz opposed, with the motors at its tips. It's damped in flight at every speed, lightly: a damping ratio of 0.02 to 0.04.

6.4 Sound

There are no piston engines: what you hear is the electric drive. Each motor whines with its own tones, and the propellers hum at their blade-passing frequencies: the two cruise propellers at the wingtips and, on Mod IV, the twelve high-lift propellers along the leading edge while they're armed. With the motors idling on the ground it's quiet, with the gear pump and flap motor clear; in flight the wind takes over, and at 100 kt it's about 10 dB over the two cruise propellers together. In flight the flap motor can't be heard in the cockpit, and the gear pump is faint. Outside, the gear out at 90 kt is about as loud as a cruise propeller. See appendix B.

Chapter 7

Emergency procedures

NASA's emergency procedures aren't public. Those here come from its papers where it says so, and otherwise are this manual's own, from what each failure does.

7.1 Motors and propellers

7.1.1 A cruise motor lost

The motor at the wingtip has a long arm. Left alone, in the model the aircraft rolls past 79 degrees in 3 seconds at 95 KCAS, with a peak roll rate of 49 degrees a second: about three times as hard as a Mod II.

WARNING

Below minimum control speed (Mod III 85 KCAS, the model 84.1) full rudder won't hold the aircraft at full power on the good side. Pull the torque lever back to cut the asymmetry.

1. Aileron and rudder against the roll, at once.
2. Good motor's torque lever back if needed to hold control.
3. Lower the nose, and hold at least 96 KIAS.
4. Failed side's propeller lever to FEATHER, to stop the dead propeller windmilling. The blades turn only 1 degree a second, so this takes over a minute, and until it's done the drag costs about 110 ft/min more.

With a motor out and feathered, gear down and takeoff flaps, Mod III at 97 KCAS sinks about 420 ft/min, so it won't climb away like that.

7.1.2 Runaway propeller

NASA's emergency stop for a runaway propeller is to pull PROP CONT. L or R, which freezes that hub.

7.1.3 High-lift motors lost (Mod IV)

If high-lift motors fail on one side, press the yoke's disarm button (Y) to remove the asymmetry, as NASA's pilots did in the simulator. RESET REQ'D lights: move ARM to DISARM/RESET.

WARNING

A contactor failure takes three discs on one wing at once, costing about 0.3 to 0.4 of the lift coefficient at 58 KCAS and rolling the aircraft. Add speed.

7.2 Electrical and battery

7.2.1 Emergency shutdown

Torque levers to idle, propeller levers to feather, MAIN BATT A and B off. Then CROSS BUS A and B, DC CONV A and B, and ESS BATT as soon as practical.

7.2.2 Annunciator alerts

Alert	Action
MAIN BATT A, B (red)	Cell above 65 °C: reduce power, land now. If it goes on rising, emergency shutdown
ESS BATT (red)	Battery failed or below 12.8 V: check the cross buses and converters. Shed AVIONICS BUS A, B. With the Essential Bus dead all traction goes
TRAC BUS A, B (yellow)	At or below 350 V: reduce power, return to base
DC CONV A, B (yellow)	Converter failed, or traction battery below 310 V: return to base
LEFT, RIGHT CMC A, B (yellow)	Controller failed, or armed with no traction voltage: that motor has half its torque or less
MAIN DOOR OPEN, REAR DOOR OPEN (red)	Land and close it

Chapter 8

Performance data

All at standard conditions. The model's figures first, NASA's in brackets. Mod III at 3000 lb for the stalls and cruise, 3075 lb for the climbs, as NASA's simulator.

Table 8.1: Rates of climb, gear down, takeoff flaps, 2500 ft, ft/min.

Mod III, 97 KCAS, full power	597 (430)
Mod III, 97 KCAS, 2400 rpm	426 (270)
Mod III, a motor out and feathered	−418 (−450)
Mod III, a traction bus out	−315 (−300)
Mod IV, DiTTo, 97 KIAS, all working	787 (720)
Mod IV, DiTTo, 97 KIAS, a motor out	86 (360)
Mod IV, DiTTo, 97 KIAS, a contactor out	599 (480)
Mod IV, DiTTo, 97 KIAS, a traction bus out	−174 (−100)
Mod IV, DiTTo, 84 KIAS, all working	1151 (1000)
Mod IV, DiTTo, 84 KIAS, a motor out	455 (530)
Mod IV, DiTTo, 84 KIAS, a contactor out	893 (740)
Mod IV, DiTTo, 84 KIAS, a traction bus out	87 (100)

Table 8.2: Other figures.

Lift-to-drag ratio at 150 KTAS, 8000 ft	14.2 (13.5)
Best lift-to-drag ratio, at 110 KCAS	15.6 (15.9)
Power for 150 KTAS, 8000 ft	82 kW
NASA's slow-flight mission, energy drawn	41.9 kWh (41.9) of 51.2 usable, in 28 minutes
Lowest voltage on that mission	378 V (370)
Hottest pack on that mission	59.3 °C (57.9)
Automated test pilot's takeoff, Mod III	lift-off after 38.9 s, 3840 ft, 90 KCAS
Automated test pilot's takeoff, Mod IV	lift-off after 12.3 s, 1190 ft, 83 KCAS

Note

The biggest gaps with NASA are the DiTTo climbs with a motor out. NASA's climb figures imply a smaller loss than its own motor tables give, and the model follows the tables. The aircraft never flew, so nobody can say which figure is right.

Appendix A

The cockpit and NASA's handbook

NASA's cockpit handbook is public: *X-57 Cockpit Interface Control Document*, ICD-CEPT-006 rev. E, at `https://ntrs.nasa.gov/citations/20230016539`. Read it beside the simulator: this cockpit follows it control by control: names, places, lights and sequences. A check confirms that each of the 119 clickable controls reaches its object and its logic: 113, and six for the Tecnam's cockpit lights. Where the handbook and this model differ, it's listed here.

A.1 Differences

- The aural alerts are one tone in place of NASA's six spoken messages, whose wording isn't published.
- The power system runs at fixed voltages, with no sag. Bus loads and the essential battery's 20 Ah are estimates.
- Motor and controller temperatures aren't modelled, so those D175 pages show dashes. Battery temperature is.
- NASA's controllers limit how fast torque changes, but the limit isn't published. The simulator takes one second for the full range, an estimate.
- Controller lights cover failures and arming without traction voltage, not internal faults.
- The cabin fan and most breakers have no effect. PITOT HEAT isn't modelled.
- The audio panel isn't modelled. The navigator shows COM 2 and NAV 1 frequencies only, and the transponder its code.
- The exterior lights work, though NASA's power drawings collar out landing, taxi and nav, the drawing with the Mod IV additions included. All four are on DC Converter Bus B in those drawings. The drawings don't say where the lights sit. Landing and taxi are in the left wing's leading edge, where NASA's photographs show them on Mod II's stock wing. No source shows them on the Mod III wing, so they keep the same distance from the centreline, which puts them behind the fourth high-lift propeller on Mod IV.
- The cockpit has no lights, as on NASA's aircraft. The P2006T's set can be fitted from the menu: section 3.5.
- FlightGear's controls can't lock a switch, so the lever locks aren't modelled.
- The gear's ground guard is a simulator choice (see Landing gear and flaps).
- The right breaker panel's order is a guess, from NASA's power drawing.
- Mod III shares this cockpit, so its high-lift controls are there but do nothing.
- The attitude, heading, turn, airspeed, altimeter and climb instruments are the stock P2006T's.

Appendix B

Sound levels

Every sound plays on one scale, so the balance between them is the physics', as far as it's known. The levels below are A-weighted, in the cockpit and 30 m outside.

Table B.1: The airframe's sounds.

Sound	Cockpit	30 m	Where the level comes from
Gear pump and jacks	75	30	estimate, from hydraulic power units' published 58 to 75 at 1 m
Gear locking down, up	85, 82.5	55, 52.5	estimate, as the ASK 21's airbrake lock
Flap motor	65	30	estimate, from electric actuators' published 55 to 70
Three tyres rolling at 60 kt	89.5	67	computed: outside within 0.5 dB of a towed light aircraft's; inside from a recorded P2006T landing, set against the wind, perhaps up to 3 dB loud. Each wheel by its own speed
A main wheel's touchdown squeal at 60 kt of slip	83	68	computed: the energy the tyre loses spinning up, by physics; the share heard and its spectrum from a recorded P2006T landing. Sound exposure, as it's a short event: a firm touchdown is shorter, as loud, and the squeal follows the wheel's load as the recorded one does. The nose wheel's is 79 and 67. A wheel still turning, after a hop, is quiet
A main strut's thump, 0.25 to 1 ft/s	69–81		estimate: barely heard in the same landing. Plays at its estimate; a firmer one is held under full scale, 3 dB under it at 2 ft/s and 7 at 4. Sound exposure
Toggle or rocker switch	78		estimate, the loudest 125 ms
Push button, breaker, knob	72		estimate, a lighter mechanism
Aural alert	105		15 dB over the cockpit's noise at 100 kt, after ISO 7731

The propellers, motors, the wind, the gear in the airflow, the tyres rolling and the touchdowns are computed by PyWopWop. The gear in the airflow is Fink's method, the same as for the ASK 21's fixed wheels: at 90 kt, 73 in the cockpit and 67 at 30 m. Nobody has heard these aircraft, so none of it has been checked against a recording of them; the tyres and touchdowns take their cockpit levels from a recorded P2006T landing. The wind (91 at 110 kt) and the motors are the least certain.

Appendix C

Sources

- X-57 Cockpit Interface Control Document, ICD-CEPT-006 rev. E: https://ntrs.nasa.gov/citations/20230016539
- Power System Human Machine Interfaces for the X-57 Mod II Aircraft: https://ntrs.nasa.gov/citations/20240006533
- X-57 Traction Power and Command Systems Development, NTRS 20240009560
- X-57 Cockpit Display System Development and Features, NTRS 20230005312
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- Development of the X-57 Mod III/IV Piloted Simulator and Discussion of the Resultant Flying Qualities Predictions, NTRS 20240006496
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